

量子力学Ⅱ 第1-3回 確認問題 解答例

1.

$$\begin{aligned}\frac{dF}{dt} &= \frac{\partial F}{\partial q} \frac{dq}{dt} + \frac{\partial F}{\partial p} \frac{dp}{dt} \\ &= \frac{\partial F}{\partial q} \frac{\partial H}{\partial p} + \frac{\partial F}{\partial p} \left(-\frac{\partial H}{\partial q} \right) \\ &= \{F, H\}\end{aligned}$$

2.

任意のケット $|\psi\rangle, |\psi'\rangle$ に対し

$$\begin{aligned}(|\psi\rangle, \hat{A}|\psi'\rangle) &= (|\psi\rangle, |\alpha\rangle\langle\beta|\psi'\rangle) \\ &= \langle\beta|\psi'\rangle(|\psi\rangle, |\alpha\rangle) \\ &= \langle\beta|\psi'\rangle\langle\psi|\alpha\rangle \\ &= \langle\psi|\alpha\rangle(|\beta\rangle, |\psi'\rangle) \\ &= (|\beta\rangle\langle\alpha|\psi\rangle, |\psi'\rangle)\end{aligned}$$

よって, $\hat{A}^\dagger = |\beta\rangle\langle\alpha|$

3.

(1)

$$\begin{aligned}\epsilon_{ijk}\epsilon_{nlm} &= |(\mathbf{e}_i\mathbf{e}_j\mathbf{e}_k)||(\mathbf{e}_n\mathbf{e}_l\mathbf{e}_m)| \\ &= |(\mathbf{e}_i\mathbf{e}_j\mathbf{e}_k)^T||(\mathbf{e}_n\mathbf{e}_l\mathbf{e}_m)| \\ &= \left| \begin{pmatrix} \mathbf{e}_i^T \\ \mathbf{e}_j^T \\ \mathbf{e}_k^T \end{pmatrix} \right| |(\mathbf{e}_n\mathbf{e}_l\mathbf{e}_m)| \\ &= \left| \begin{pmatrix} \mathbf{e}_i^T \\ \mathbf{e}_j^T \\ \mathbf{e}_k^T \end{pmatrix} (\mathbf{e}_n\mathbf{e}_l\mathbf{e}_m) \right| \\ &= \left| \begin{pmatrix} \mathbf{e}_i^T\mathbf{e}_n & \mathbf{e}_i^T\mathbf{e}_l & \mathbf{e}_i^T\mathbf{e}_m \\ \mathbf{e}_j^T\mathbf{e}_n & \mathbf{e}_j^T\mathbf{e}_l & \mathbf{e}_j^T\mathbf{e}_m \\ \mathbf{e}_k^T\mathbf{e}_n & \mathbf{e}_k^T\mathbf{e}_l & \mathbf{e}_k^T\mathbf{e}_m \end{pmatrix} \right| \\ &= \left| \begin{pmatrix} \delta_{in} & \delta_{il} & \delta_{im} \\ \delta_{jn} & \delta_{jl} & \delta_{jm} \\ \delta_{kn} & \delta_{kl} & \delta_{km} \end{pmatrix} \right| \\ &= \delta_{in}\delta_{jl}\delta_{km} + \delta_{il}\delta_{jm}\delta_{kn} + \delta_{im}\delta_{jn}\delta_{kl} \\ &\quad - \delta_{im}\delta_{jl}\delta_{kn} - \delta_{il}\delta_{jn}\delta_{km} - \delta_{in}\delta_{jm}\delta_{kl}\end{aligned}$$

(2) $n = i$ とおくと

$$\begin{aligned}\epsilon_{ijk}\epsilon_{ilm} &= \delta_{ii}\delta_{jl}\delta_{km} + \delta_{il}\delta_{jm}\delta_{ki} + \delta_{im}\delta_{ji}\delta_{kl} \\ &\quad - \delta_{im}\delta_{jl}\delta_{ki} - \delta_{il}\delta_{ji}\delta_{km} - \delta_{ii}\delta_{jm}\delta_{kl} \\ &= 3\delta_{jl}\delta_{km} + \delta_{jm}\delta_{kl} + \delta_{jm}\delta_{kl} \\ &\quad - \delta_{jl}\delta_{km} - \delta_{jl}\delta_{km} - 3\delta_{jm}\delta_{kl} \\ &= \delta_{jl}\delta_{km} - \delta_{jm}\delta_{kl}\end{aligned}$$